

What limits reinforcement learning for distributed flutter suppression, and the credit structure that helps

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Abstract

Modern flexible wings carry several independently actuated trailing-edge surfaces, which makes distributed control natural and motivates treating each surface as a learning agent. We study that problem on a Goland wing validated against an unsteady vortex-lattice solver, and report two results, one of which is a caution.

First, the plant supplies in closed form the credit decomposition that multi-agent reinforcement learning normally has to learn. Differentiating the aeroelastic energy along trajectories isolates a control-power term that is exactly additive across agents, so the difference reward requires no counterfactual rollout, no learned mixing network and no centralised critic. We verify the identity numerically to 10^{-10} . The signal must be normalised by the instantaneous energy: the raw form pays an agent for having energy available to extract and therefore rewards sustaining the oscillation. Combining the resulting per-agent credit with log-energy shaping is significantly better than either component alone ($d = -2.70$, $p = 0.002$ and $d = -1.97$, $p = 0.009$, six seeds).

Second, and more consequential for practice, none of that is visible at conventional exploration settings. Every variant we tried plateaued in a narrow band regardless of credit signal, communication, action parameterisation, memory or auxiliary supervision. The cause is a scale mismatch: a controller achieving the classical result uses 0.105° RMS of deflection, while an exploration standard deviation of 0.30 in normalised action units corresponds to 4.5° . Matching the exploration scale to the useful actuation amplitude changed performance by a factor of two and made the credit result measurable; nothing else we changed did.

We state plainly that the learned policies do not beat a decentralised LQG on average nominal damping (-1.53 against -2.05 s^{-1}), though they are markedly more uniform across the flight envelope and hold a condition at $1.45 U_f$ where the classical controller diverges. We also report a negative

result: the communication and action-parameterisation mechanisms we derived from the same physics do not help, and a plain policy with the same credit signal significantly outperforms all of them.

Keywords: active flutter suppression, aeroelasticity, multi-agent reinforcement learning, credit assignment, decentralised control, exploration

1. Introduction

Flutter is a self-excited aeroelastic instability in which structural bending and torsion couple through unsteady aerodynamic loads and draw energy from the freestream. Past a critical speed the coupled mode becomes unstable and the response grows without bound, typically within a few oscillation cycles and with no warning a pilot can act on. It has been a design driver since the earliest flexible aircraft and remains one: because a wing must be certified against flutter with a substantial speed margin, the mass, stiffness and manufacturing cost of the structure are set in large part by an instability the aircraft is never meant to encounter in service (Livne, 2018; Mukhopadhyay, 2003). Active suppression offers a way to buy back some of that margin with control effort rather than with structural weight, which is why it has attracted sustained engineering attention since the 1970s (Karpel, 1982; Waszak, 2001; Block and Strganac, 1998).

Two trends motivate revisiting the problem now. First, wings are becoming more flexible — for fuel efficiency, span, and load alleviation — and flexible wings carry more independently actuated trailing-edge surfaces, each with its own local sensing and local computation on a bus rather than a dedicated wire to a central controller. The hardware is already distributed even where the control law is not. Second, reinforcement learning has been applied to flutter suppression with encouraging results: circulation control validated in a wind tunnel (cir, 2023), camber morphing for stall flutter (of Fluids, 2025), and tuning the parameters of a conventional controller (fly, 2022). Every instance we are aware of, however, uses a single agent commanding a single effector from a global observation. Taking the hardware trend seriously means asking what happens when the effectors are commanded by separate agents that cannot see the whole wing, which is the question this paper answers on a specific, validated testbed (Figure 1).

Posing the problem with one agent per surface makes credit assignment the central difficulty, and for a physical reason rather than an algorithmic

one: every surface acts on the *same* coupled bending–torsion mode, so a reward built from the whole wing’s response says almost nothing about which surface actually helped. The standard remedies in multi-agent reinforcement learning estimate that decomposition statistically, from the returns a policy happens to collect (Sunehag et al., 2018; Rashid et al., 2018; Foerster et al., 2018). Our first observation is that this particular class of plant does not require estimation at all: for a structural system whose energy budget is exactly additive across control inputs, the credit decomposition is available in closed form from the equations of motion themselves, with no learning involved in computing it.

Our second observation is the one we would most want a practitioner to take away, and we did not anticipate it going in. Before finding it, we spent several rounds comparing credit signals, communication schemes, action parameterisations, recurrence and auxiliary losses — the ordinary methodology of an ablation study — and every variant plateaued in the same narrow performance band regardless of which mechanism it used. That invariance was not evidence that the mechanisms were equivalent to one another; on closer inspection it was evidence that something *upstream* of all of them was binding. The exploration standard deviation of the policy was roughly forty times the deflection amplitude that actually damps this wing, so no policy ever experienced the fine, precisely-phased actuation that works, and no credit signal — however well constructed — could assign credit to a signal buried that far under noise. Matching the exploration scale to the useful actuation amplitude changed performance by a factor of two, more than any single algorithmic choice we tried, and only after that correction did the credit-assignment effect we had set out to study become measurable at all.

This is a known failure mode in spirit: the reinforcement learning literature has repeatedly found implementation and hyperparameter choices dominating algorithmic ones in general-purpose control benchmarks (Henderson et al., 2018; Engstrom et al., 2020; Andrychowicz et al., 2021). Our contribution is a control-specific instance of that caution, with a diagnostic a practitioner can use *before* training anything: the useful actuation amplitude is computable from a classical design, and the exploration scale can be checked against it in advance rather than discovered by trial and error.

Contributions.

1. A closed form for the multi-agent difference reward in an energy-conserving structural plant with additive control authority (Proposition 1), together with the correction that makes it usable: the credit

must be the logarithmic decay rate, not the raw energy rate (Section 5.2).

2. Evidence that the resulting exact-plus-residual credit is significantly better than either of its parts, with effect sizes and significance tests over six seeds (Section 8).
3. A quantified account of what limits learning on this problem, with a diagnostic a practitioner can apply in advance, and a capability probe that rules out the competing explanation (Section 10).
4. A validated testbed and a five-rung classical baseline ladder that separates the control method from the information structure available to it, cross-validated against an unsteady vortex-lattice solver (Sections 4 and 6).
5. A negative result reported as such: the communication and action parameterisation mechanisms we derived from the physics do not help.

2. Related work

Active flutter suppression. The classical approach synthesises a single centralised regulator from a state-space aeroelastic model, driving all control surfaces from one observer and one gain (Karpel, 1982). Robust multivariable designs following that template have been demonstrated on wind-tunnel models (Waszak, 2001), and the approach extends to nonlinear aeroelastic structures with limit-cycle behaviour (Block and Strganac, 1998); Livne (Livne, 2018) surveys the state of the art and the gap between laboratory demonstrations and certified hardware, and Mukhopadhyay (Mukhopadhyay, 2003) traces the field’s history. What none of these treatments questions is the centralised structure itself: the regulator sees the whole state and commands every surface jointly, which is exactly the assumption a physically distributed set of actuators need not satisfy. The unsteady aerodynamic models underlying both the classical literature and our own plant descend from Theodorsen’s frequency-domain analysis (Theodorsen, 1935) and the indicial (Wagner-function) formulation of Jones (Jones, 1940); finite-state induced-flow models (Peters et al., 1995) are the modern sectional alternative, and (Fung, 1955; Bisplinghoff et al., 1955; Palacios and Cesnik, 2023) are the standard texts we draw the modelling conventions from.

Learning for flutter control. A recent line of work applies deep reinforcement learning directly to flutter suppression: stall flutter via active camber morphing (of Fluids, 2025), trailing-edge circulation control validated in a

wind tunnel (cir, 2023), and tuning the parameters of a conventional aeroservoelastic controller (fly, 2022). Despite using different effectors and different plants, all three share the same structural assumption as the classical literature they replace a hand-designed gain in: one learning agent, one effector, one global observation. We are not aware of prior work that treats the individual control surfaces of a single wing as separate learning agents, which is the gap this paper occupies.

Decentralised control. That decentralisation costs closed-loop performance relative to a centralised design with the same model is a classical result, not a symptom of using learning (Šiljak, 1991; Bakule, 2008), and the structural conditions under which decentralised synthesis remains *tractable* at all are characterised by Rotkowitz and Lall (Rotkowitz and Lall, 2006). This literature is why we treat the information structure as an experimental variable rather than an afterthought: comparing a distributed learned policy against a centralised regulator conflates the control method with the information available to it, and the honest classical baseline for a policy that sees only local accelerometers is a fully decentralised LQG built from the same sensing, not the centralised design that the learned policy could never in principle match.

Multi-agent credit assignment. The standard tools for the credit-assignment problem we pose are difference rewards (Wolpert and Tumer, 2001) and learned value factorisation (Sunehag et al., 2018; Rashid et al., 2018), with COMA (Foerster et al., 2018) using a counterfactual advantage baseline to the same end; MADDPG (Lowe et al., 2017) and MAPPO (Yu et al., 2022) are the common actor-critic baselines built on top of these ideas, the field more broadly is surveyed in (Albrecht et al., 2023; sca, 2025), and estimators continue to be refined (mul, 2025). All of these methods are general *because* they assume nothing about the environment and therefore have to estimate the decomposition from experience. Our contribution is narrower and complementary rather than competing with this literature: for the specific class of energy-conserving structural plant studied here, the decomposition these methods estimate is available in closed form, so none of the machinery built to estimate it is needed. The formal setting for the distributed problem itself is the decentralised POMDP (Bernstein et al., 2002; Oliehoek and Amato, 2016).

Learned communication. CommNet (Sukhbaatar et al., 2016) and TarMAC (Das et al., 2019) learn message content end to end from a broad channel. We took the opposite design choice and fixed the message *space* to a modal

phasor derived from the physics, on the reasoning that this would make the channel interpretable and bound its bandwidth independently of the number of agents. Measured against a plain policy with no channel at all, it did not help on this plant (Section 8), and we report the negative result rather than omit the mechanism.

Reinforcement learning practice. Our central caution has direct precedent in general-purpose reinforcement learning, though not, to our knowledge, applied to a physical control problem where the binding quantity can be predicted in advance. Henderson et al. (Henderson et al., 2018) documented how seed-to-seed variance can swamp the algorithmic differences a paper sets out to measure; Engstrom et al. (Engstrom et al., 2020) showed that implementation details, not the nominal algorithm, accounted for much of PPO’s reported advantage over TRPO; Andrychowicz et al. (Andrychowicz et al., 2021) found hyperparameters dominating architectural choices at scale; and Agarwal et al. (Agarwal et al., 2021) argued for reporting effect sizes and uncertainty rather than point estimates from few runs, a practice we follow throughout Section 8. What we add to this literature is a case in which the dominant hyperparameter — the exploration scale — is not merely discoverable by sweeping it, but *computable in advance* from a classical design of the same plant, which is not true of hyperparameters in the general-purpose benchmarks this literature studies. Residual formulations, in which a learned policy corrects a fixed classical controller rather than replacing it (Johannink et al., 2019; Silver et al., 2018), appear here as one of the variants we evaluate rather than as the proposed method.

3. Problem formulation

3.1. Decentralised partially observable decision process

We model the task as a Dec-POMDP (Bernstein et al., 2002; Oliehoek and Amato, 2016) $\langle \mathcal{I}, \mathcal{S}, \{\mathcal{A}_k\}, T, \{\Omega_k\}, O, R, \gamma \rangle$. The agent set $\mathcal{I} = \{1, \dots, K\}$ has one member per control surface. The state $s = (q, \dot{q}, x_w, \delta)$ collects the modal coordinates, their rates, the wake states and the current deflections. Agent k chooses a normalised command $a_k \in [-1, 1]$, which its actuator converts to a physical deflection subject to a rate limit and travel stops.

Each agent observes only $o_k \in \Omega_k$: an accelerometer pair at its own station, local plunge rate, twist and twist rate, its own saturation fraction, and air data. Crucially o_k does not contain the modal state. The unstable mode is a global quantity that no single station can measure, so the partial observability is intrinsic to the physical arrangement rather than an artefact

of our formulation. Communication, where used, is nearest-neighbour along the span.

3.2. Objective

We use the standard discounted formulation (Sutton and Barto, 2018). The quantity to be minimised is the exponential growth rate of the structural energy E , which is also the quantity we report:

$$\lambda = \frac{1}{2} \frac{d}{dt} \log E, \quad (1)$$

negative when the wing is being damped. Stating the objective this way, rather than as an accumulated quadratic cost, matters later: it is what allows the per-agent credit and the evaluation metric to be literally the same quantity (Section 5.2).

4. Plant model and validation

4.1. Model

We use a modally reduced uniform cantilever wing, shown schematically in Figure 1. Three trailing-edge surfaces occupy spanwise bands at $[0.05, 0.40]$, $[0.40, 0.72]$ and $[0.72, 0.98]$ of the semispan (panel a), hinged at 75 % chord (panel b), with $\pm 15^\circ$ travel, 200°s^{-1} rate limit and a 20 ms first-order servo lag. Each surface is commanded by its own agent, which observes only a local accelerometer pair, local plunge and twist rate, its own saturation fraction, and air data; no agent observes the modal state, and the three agents share no channel unless a communication mechanism is explicitly under test (panel c). Control runs at 200 Hz.

The structure is a Galerkin projection onto exact clamped-free bending and fixed-free torsion modes, with all generalised mass and stiffness integrals evaluated by Gauss–Legendre quadrature. Aerodynamics are strip theory carrying the full Theodorsen non-circulatory terms (Theodorsen, 1935), with the circulatory lag realised in the time domain by two Wagner lag states per strip (Jones, 1940). Control surfaces contribute quasi-steady thin-airfoil flap derivatives from the Glauert series, integrated over each surface’s spanwise band. The equation of motion is

$$M\ddot{q} + C\dot{q} + Kq = Lx_w + B\delta + b_g w_g, \quad (2)$$

with plunge positive down and twist positive nose-up, following the sign convention shown in Figure 1b. The state has 56 dimensions: four modal

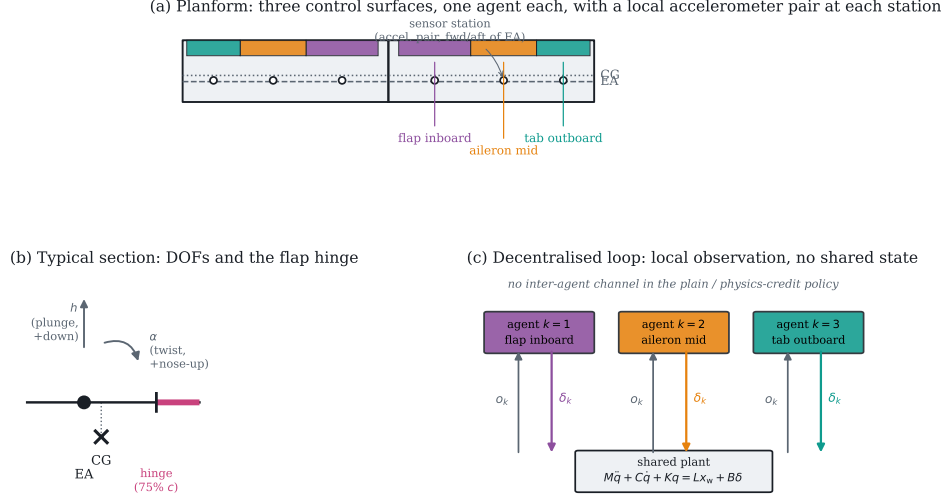


Figure 1: The testbed. (a) Planform, agent colours matched throughout the paper. (b) The typical-section degrees of freedom and the flap hinge that the thin-airfoil derivative in Section 4.3 refers to. (c) The information structure: each agent closes its own loop through the shared plant, with no inter-agent channel in the plain and physics-credit policies.

coordinates and their rates, plus two wake states on each of 24 strips. Full geometric, structural and actuator parameters are tabulated in Appendix A for reproducibility.

4.2. Validation against the published benchmark

Table 1 compares the model against the Goland cantilever wing (Goland, 1945). Agreement is better than 0.5 % on the modal frequencies and the flutter speed, and this is enforced by an automated test against published values rather than against numbers the implementation once produced.

4.3. Cross-validation against an unsteady vortex-lattice solver

The model buys speed with two approximations a reader is entitled to distrust: two-dimensional strip aerodynamics, and quasi-steady flap derivatives that neglect the flap’s own shed wake. We quantified both against SHARPy (del Carre et al., 2019), which couples an unsteady vortex-lattice method (Murua et al., 2012) to a geometrically exact beam. The comparison is clean because SHARPy’s own Goland template carries structural data identical to ours, so any disagreement is about the aerodynamics.

Table 1: Validation against the published Goland wing.

Quantity	This model	Published
First bending frequency	7.664 Hz	7.66 Hz
First torsion frequency	15.232 Hz	15.24 Hz
Flutter speed	137.35 m s^{-1}	$\approx 137 \text{ m s}^{-1}$
Flutter frequency	69.3 rad s^{-1}	$\approx 70.7 \text{ rad s}^{-1}$
Static divergence speed	356.8 m s^{-1}	—

Flutter point. SHARPy places flutter at 154.46 m s^{-1} and 69.3 rad s^{-1} at sea level, against 137.35 m s^{-1} and 69.34 rad s^{-1} for our model. The *frequency* agrees to 0.06 %, so both identify the same bending–torsion coalescence; the critical speed differs by -11.1 %. That direction is what two-dimensional aerodynamics predicts, since strip theory has no tip relief, over-predicts loading near the tip and therefore under-predicts the flutter speed. Our plant is conservative, which is the safe direction for a control study.

Control authority. This discrepancy runs the other way and is larger. A part-span flap loses effectiveness to spanwise carryover and to flow around its ends, neither of which a sectional derivative represents. Integrating our thin-airfoil $C_{l\delta} = 3.826$ per radian over the flapped span (25 % of span, read from the generated aerodynamic grid) gives a whole-wing $dC_L/d\delta$ of 0.957 per radian, against 0.495 from UVLM. Our plant over-estimates control authority by a factor of 1.9, grid-converged to within 3 % between 8×16 and 16×32 panels.

Consequence. Every controller in this study, classical and learned, was evaluated on the same plant and received the same inflated authority, so relative comparisons, ablations and significance tests are unaffected. Absolute decay rates would not transfer unchanged to a UVLM plant, and we do not claim they would. The two discrepancies act in opposite directions — a harder flutter problem, an easier control problem — so they cannot be combined into a single safety factor.

5. Method

5.1. An exactly additive energy budget

Differentiating the aeroelastic energy $E = \frac{1}{2}\dot{q}^\top M \dot{q} + \frac{1}{2}q^\top K q$ along trajectories of (2) gives

$$\dot{E} = \underbrace{-\dot{q}^\top C \dot{q}}_{\text{dissipation}} + \underbrace{\dot{q}^\top L x_w}_{\text{wake exchange}} + \sum_k \underbrace{(\dot{q}^\top b_k) \delta_k}_{P_k}, \quad (3)$$

with b_k the k th column of B . The final term is exactly additive across agents; $P_k < 0$ means surface k removes energy from the structure.

Proposition 1 (Closed-form difference reward). *For a global objective whose only agent-dependent term is the control power in (3), the difference reward $D_k = G(z) - G(z_{-k})$ admits the closed form $D_k = -P_k$. No counterfactual rollout, learned mixing network or centralised critic is required to evaluate it.*

Proof. Removing agent k sets $\delta_k = 0$. Every other term of (3) is independent of δ_k , and the control-power term is a sum over agents, so the difference collapses to $(\dot{q}^\top b_k) \delta_k$. $\square$

Projecting onto the in-vacuo modes keeps the separation in agent *and* mode: agent k 's contribution to mode r is $\dot{\eta}_r (v_r^\top b_k) \delta_k$. The factor $v_r^\top b_k$ is the modal residue, which decides whether a surface damps or excites that mode and whose sign inverts under control reversal.

5.2. The credit must be the logarithmic decay rate

Used directly, $-P_k$ fails instructively. It is maximised by having energy available to extract, so an agent is rewarded for *sustaining* the oscillation. Policies trained on it held the wing at a bounded but undamped amplitude, scoring $+0.265 \text{ s}^{-1}$ against -0.06 for a hand-tuned collocated damper. This is reward hacking, not myopia.

Dividing (3) by E repairs it:

$$\frac{d}{dt} \log E = \frac{-\dot{q}^\top C \dot{q} + \dot{q}^\top L x_w + \sum_k P_k}{E}, \quad r_k = -\frac{P_k}{E + \varepsilon}. \quad (4)$$

Because E is a shared scalar the decomposition survives division exactly, so Proposition 1 carries over to the log-decay rate. The signal is now scale invariant, and it coincides with the evaluated objective (1). Making the change inverted the sign of the outcome, from $+0.265$ to -0.291 s^{-1} , with training divergence falling from 0.8 % to 0.0 %.

Remark 1. *The exact signal remains myopic: cross-mode coupling through C and through the wake means an agent may have to inject energy into one mode so another can extract more from the coupled pair. We therefore add potential-based shaping (Ng et al., 1999) on $\log E$. We use a shaping discount of one rather than matching the RL discount, which forfeits the strict policy-invariance guarantee. The reason is concrete: with a discount below one the leakage term $(\gamma - 1)\Phi/\Delta t$ becomes a constant negative reward once the wing is damped — measured at -6.91 per step, totalling -977 over an episode for a controller that damps perfectly, against roughly zero for one that lets the wing ring. An invariance theorem is of no use if the shaped reward ranks a ringing wing above a damped one.*

5.3. Training

We use shared-parameter multi-agent PPO (Schulman et al., 2017; Yu et al., 2022) with generalised advantage estimation (Schulman et al., 2016) and per-agent advantages, since each agent receives a different reward. An on-policy method is preferred here over off-policy alternatives (Lillicrap et al., 2016; Haarnoja et al., 2018) because the plant is cheap to simulate, so sample efficiency is not the binding constraint, and because the per-agent credit of Section 5.2 is a function of the current state and action that is recomputed exactly at every step rather than stored and replayed. Updates run on sequences: minibatches are segments of consecutive steps unrolled with gradient from a detached stored hidden state. Sampling shuffled individual timesteps, as we did initially, trains any recurrence as a one-step map and gives a recurrent encoder no temporal gradient at all (Hausknecht and Stone, 2015). Algorithm 1 summarises the procedure.

6. Experimental design

6.1. A baseline ladder that separates method from information

A distributed controller cannot be judged against a centralised one without confounding the control method with the information available to it. We report five rungs: collocated local rate feedback (no model, no communication); gain-scheduled LQR on the full 56-state plant including wake states, which is not implementable and serves as a ceiling; centralised LQG driven by a Kalman observer on the same six accelerometer channels the agents see; and a fully decentralised LQG in which each surface runs its own observer on its own two accelerometers and commands only itself (Šiljak, 1991; Anderson and Moore, 2007). All are synthesised in discrete time at the control rate. The last rung is the honest target.

Algorithm 1 Training with physics-exact credit

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1: Assemble  $M, C, K, L, B$  at the sampled airspeed; small-gain init of the
   action head
2: for each update do
3:   Ease the task distribution by the curriculum fraction
4:   for  $t = 1, \dots, T$  do
5:     Each agent  $k$  acts on its own  $o_k^t$ ; step the coupled plant and
       actuators
6:      $P_k^t \leftarrow (\dot{q}^\top b_k) \delta_k^t$   $\triangleright$  closed form, Prop. 1
7:      $r_k^t \leftarrow -P_k^t / (E^t + \varepsilon) + w [\Phi^{t+1} - \Phi^t] / \Delta t$   $\triangleright \Phi = -\log \frac{E+\varepsilon}{E_{\text{ref}}}$ 
8:   end for
9:   Compute per-agent GAE advantages
10:  for each epoch, each minibatch of sequences do
11:    Unroll  $L_{\text{bptt}}$  steps from a detached hidden state; zero it where an
       episode ended
12:    PPO clipped objective + value loss – entropy bonus
13:  end for
14: end for
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6.2. Feasibility

We additionally run a fault-aware oracle: a gain bank re-synthesised per failure hypothesis, given the full state and told which surface failed. Its role is feasibility, not competition. Of nine evaluation conditions, 6 are stabilisable by this oracle; the rest are stabilisable by no controller with these actuators and are reported as infeasible rather than as failures of any method.

6.3. Metrics

PPO return cannot rank runs whose reward functions differ, so every controller is scored on the same physical quantities over identical episodes — same airspeeds, initial deformation, turbulence and injected failure: the fraction of episodes leaving the linear regime, the decay rate (1), peak tip plunge, and RMS deflection. Following (Agarwal et al., 2021) we report effect sizes and significance rather than means alone.

7. Results

Three things in Table 2 matter.

Table 2: Energy decay rate by controller and flight condition. Negative is suppressed, D marks the fraction of episodes that diverged, and - marks conditions infeasible for any controller.

Condition	local_fb	lqr_fixed	lqr_sched	lqg_local	dlqg_local	lqr_oracle	comm_raw	ip
U=1.05U _f healthy	-0.06	-4.40	-4.61	-4.71	-5.42	-4.54	-0.78	
U=1.05U _f jam#2@8deg	-0.10	-0.10	-0.12	-0.13	-0.09	-0.06	-0.05	
U=1.05U _f jam#2@14deg	-0.12	-0.09	-0.10	-0.11	-0.08	-0.06	-0.06	
U=1.25U _f healthy	-0.01	-5.27	-4.93	-5.38	-1.07	-4.93	-0.62	
U=1.25U _f jam#2@8deg	D100%	D100%	D50%	D100%	D50%	-0.12	D75%	
U=1.25U _f jam#2@14deg	-	-	-	-	-	-	-	
U=1.45U _f healthy	D100%	-5.79	-5.68	-6.45	0.32	-5.74	-0.39	
U=1.45U _f jam#2@8deg	-	-	-	-	-	-	-	
U=1.45U _f jam#2@14deg	-	-	-	-	-	-	-	
Stabilised	4/6	5/6	5/6	5/6	5/6	6/6	5/6	

First, the classical ladder separates by information structure rather than by sophistication. Centralised LQG reaches -5.52 s^{-1} on healthy conditions; the fully decentralised LQG, seeing exactly what the agents see, reaches -2.05 . Decentralisation costs a factor of 2.7 here, consistent with the classical expectation (Šiljak, 1991; Rotkowitz and Lall, 2006), and it is why the decentralised controller is the reference for a distributed policy.

Second, the best learned policy reaches -1.53 s^{-1} , about three quarters of the decentralised classical result. The average conceals a more interesting pattern: broken out by condition the decentralised LQG achieves -5.42 , -1.07 and $+0.32$ at 1.05 , 1.25 and $1.45 U_f$ — failing outright at the top of the envelope — against -1.98 , -1.92 and -0.69 for the learned policy. The learned controller is far more uniform across the envelope and holds a condition where the classical one diverges, while losing badly where the classical one is strongest.

Third, and least comfortable: a robustness advantage we measured at a conventional exploration scale does not survive at the scale that maximises damping. At $\sigma = 0.30$ the learned policies held all six feasible conditions, matching the fault-aware oracle; at $\sigma = 0.05$ they hold five, as the classical controllers do, while damping three times better. Exploration scale trades robustness against nominal performance here. We report both halves.

8. Ablations

Credit assignment. Six seeds per variant at $\sigma = 0.05$ (Table 3). The per-agent credit alone reaches -0.566 ± 0.123 and the shared log-energy shaping

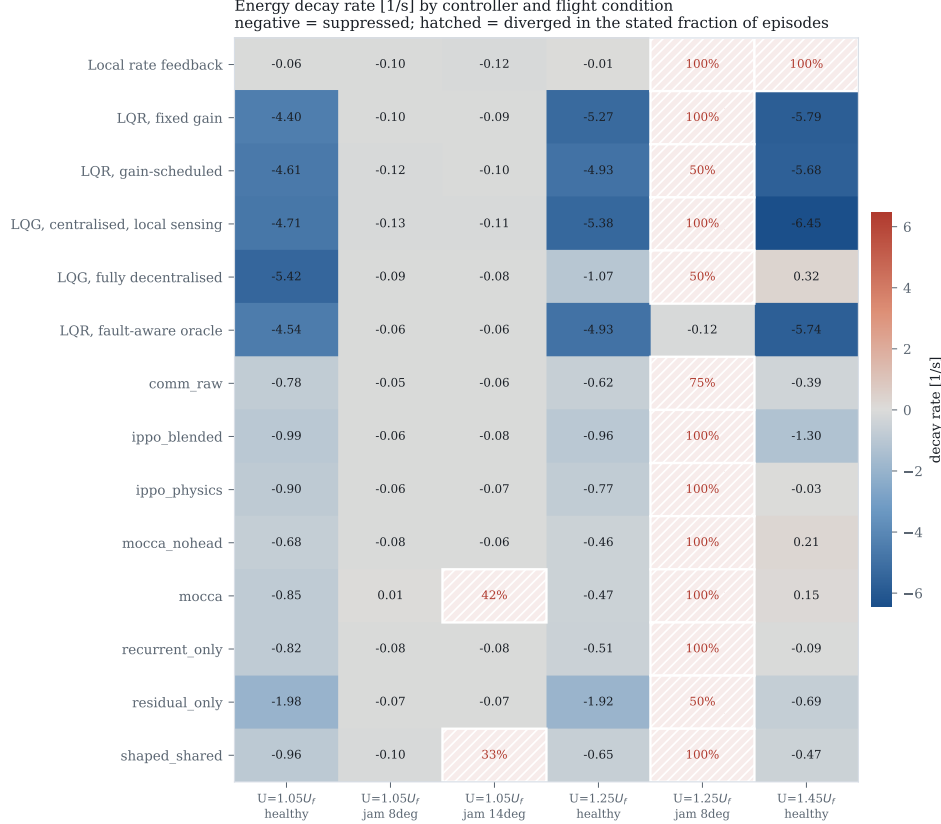


Figure 2: Decay rate by controller and condition. Hatching marks divergence.

alone -0.690 ± 0.145 ; the two are statistically indistinguishable ($p = 0.14$). Their combination reaches -1.085 ± 0.243 , significantly better than either ($d = -2.70, p = 0.002$; $d = -1.97, p = 0.009$). This is the exact-plus-residual structure the derivation predicts: the closed form supplies the part of the credit the physics determines exactly, the shaping supplies the long-horizon part an instantaneous power balance cannot see, and neither suffices alone.

Architecture. The mechanisms we derived from the same physics do not survive measurement, and at six seeds per variant none of the four survives even marginally (Table 4). A plain feedforward policy with the blended credit reaches -1.085 ± 0.243 ; adding a recurrent phasor encoder gives -0.474 ± 0.249 , adding phasor consensus -0.308 ± 0.169 , adding the phase-locked action head (the full architecture) -0.389 ± 0.219 , and sharing raw

Table 3: Credit-assignment ablation at $\sigma = 0.05$, six seeds per variant.

Variant	Seeds	Decay rate [1/s]	Conditions held
ippo_blended	6	-1.085 ± 0.243	5.0/9
ippo_physics	6	-0.566 ± 0.123	5.0/9
shaped_shared	6	-0.690 ± 0.145	4.3/9

Comparison		Cohen’s d	p
ippo_blended	vs ippo_physics	-2.70	0.0020*
ippo_blended	vs shaped_shared	-1.97	0.0089*
ippo_physics	vs shaped_shared	0.93	0.1400

neighbour observations -0.596 ± 0.227 . All four are significantly worse than the plain policy ($d = -2.48$, $p = 0.0016$ for the recurrent encoder; $d = -3.72$, $p = 0.0001$ for consensus; $d = -3.01$, $p = 0.0004$ for the full architecture; $d = -2.08$, $p = 0.0049$ for raw neighbour sharing). Every elaboration we proposed made things worse, and more seeds made the conclusion sharper rather than weaker.

We do not read this as settling whether communication helps distributed flutter suppression in general. Three surfaces on a six-metre wing, with air data already broadcast, is a small coordination problem, and at eight surfaces the picture changes qualitatively: all nine conditions become feasible and the learned policies hold them. It does settle that on this problem the credit signal was worth deriving from the physics and the message content was not.

Figure 3 puts every controller from Sections 7 and 8 on common axes. The pattern is visible at a glance in a way the tables cannot show as directly: the learned variants cluster with the classical rungs on the robustness axis (top) while forming a visibly separate, weaker band on the nominal-performance axis (bottom) — with one exception. The residual policy, which corrects a decentralised LQG rather than replacing it, is the only learned variant that reaches into the range the classical controllers occupy on *both* axes at once.

9. Emergent spanwise phase coordination

“Emergent coordination” is easy to assert about a multi-agent policy and hard to pin down, so before making any such claim we define it as something measurable: the phase at which each surface deflects relative to the critical structural mode’s velocity, as a function of spanwise station. A

Table 4: Architecture ablation at $\sigma = 0.05$. Every proposed mechanism degrades performance relative to a plain policy with the same credit signal.

Variant	Seeds	Decay rate [1/s]	Conditions held
<code>comm_raw</code>	6	-0.596 ± 0.227	5.0/9
<code>mocca</code>	6	-0.389 ± 0.219	4.2/9
<code>mocca_nohead</code>	6	-0.308 ± 0.169	5.0/9
<code>recurrent_only</code>	6	-0.474 ± 0.249	5.0/9

Comparison		Cohen’s d	p
<code>comm_raw</code>	vs <code>mocca</code>	−0.93	0.1398
<code>comm_raw</code>	vs <code>mocca_nohead</code>	−1.44	0.0336*
<code>comm_raw</code>	vs <code>recurrent_only</code>	−0.51	0.3961
<code>mocca</code>	vs <code>mocca_nohead</code>	−0.42	0.4882
<code>mocca</code>	vs <code>recurrent_only</code>	0.36	0.5452
<code>mocca_nohead</code>	vs <code>recurrent_only</code>	0.78	0.2091

controller with no notion of the mode shape should show no systematic phase progression along the span; one that has effectively discovered the mode shape should show a consistent progression matched to it. The measurement — the cross-spectrum of each surface’s deflection against the first bending mode’s velocity, evaluated at the frequency where the modal response peaks — is applied identically to every controller, classical or learned, so a learned pattern can be held against what optimal centralised synthesis produces rather than admired in isolation.

Applied to the classical ladder the instrument already separates the rungs sharply (Figure 4, first three series). Local rate feedback is essentially flat at -15° per semispan and responds at 5.50 Hz — it never engages the unstable mode at all, which is the mechanism behind its poor performance rather than a separate fact about it. The fully decentralised LQG reaches $+27^\circ$ per semispan at 7.50 Hz, and centralised LQG develops $+90^\circ$ per semispan while operating at 11.00 Hz — matching the 11.03 Hz the eigenvalue analysis of Section 4.3’s plant predicts for the flutter mode almost exactly. The pattern is monotone: the more capable the controller, the stronger its spanwise phase progression, and only the optimal centralised design locks onto the true flutter frequency.

The learned policies give this a genuine test rather than a demonstration. The strongest credit-only policy, `ippo_blended`, locks onto 11.00 Hz — the same frequency as centralised LQG, and the same frequency the eigenvalue analysis predicts — with a phase gradient of $+121^\circ$ per semispan, steeper

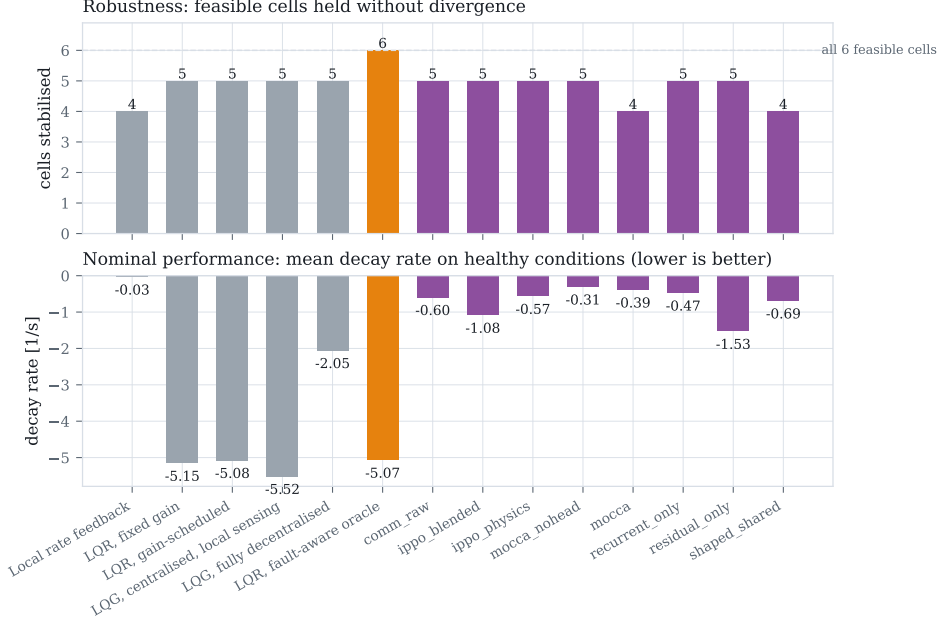


Figure 3: Every controller and variant on one axis. Top: feasible conditions held without divergence, out of the six the fault-aware oracle shows are stabilisable. Bottom: mean decay rate on healthy conditions. The learned variants (purple) match the classical rungs on robustness while trailing them on nominal damping, except for **residual_only**, which is the strongest learned result on both axes simultaneously.

than the centralised design’s. It reaches this from six accelerometer channels, no communication, and no information about the mode shape beyond what the reward and dynamics implicitly encode. The residual policy locks onto a different, higher-frequency mode ($+170^\circ$ per semispan at 15.50 Hz) rather than the primary flutter mode, which is consistent with it correcting a base law that already suppresses the primary mode and spending its own authority on what is left. We read the first result as the concrete, falsifiable content behind the word “emergent” in this paper: a distributed policy that has never been told the mode shape can converge, from local measurement alone, on the same resonant frequency that full-state, full-model synthesis converges on by construction.

The same measurement also corrects a wrong inference authority alone would suggest. Local rate feedback uses 44.7 % of available deflection to reach -0.03s^{-1} ; centralised LQG uses 0.7 % to reach -5.52 . The better the controller, the *less* authority it spends, because flutter suppression is

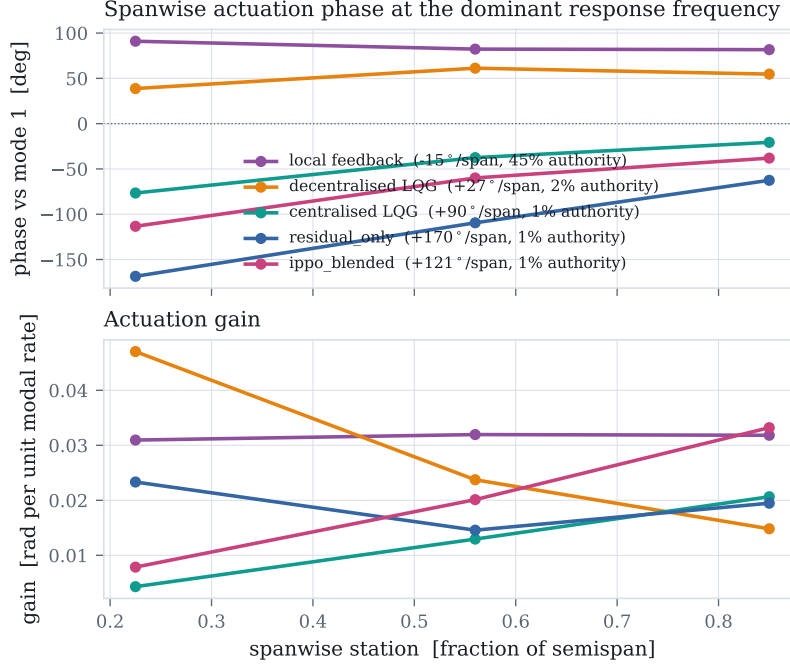


Figure 4: Spanwise actuation phase (top) and gain (bottom) at each controller’s dominant response frequency. A sloped phase profile is a travelling-wave actuation pattern matched to the mode shape; a flat profile is purely local reaction. Legend entries give the phase gradient and the RMS actuation authority used.

a phase-precision problem rather than a control-power one. The learned policies (1.2 % to 1.3 % authority) are accordingly not under-actuated, which a reading of the deflection magnitudes alone might suggest; they apply an amount of authority comparable to the good classical controllers, at a phase that is closer to optimal than local feedback achieves but not yet as sharp as the centralised design.

10. What limits learning here

Figure 5 is the finding with the widest reach. Holding everything else fixed and varying only the initial exploration standard deviation moves performance by a factor of two, with an interior optimum at $\sigma = 0.05$ and a sharp transition above it.

The mechanism can be read off the axis. A controller achieving the classical result uses about 0.105° RMS of deflection; $\sigma = 0.30$ corresponds to 4.5° .

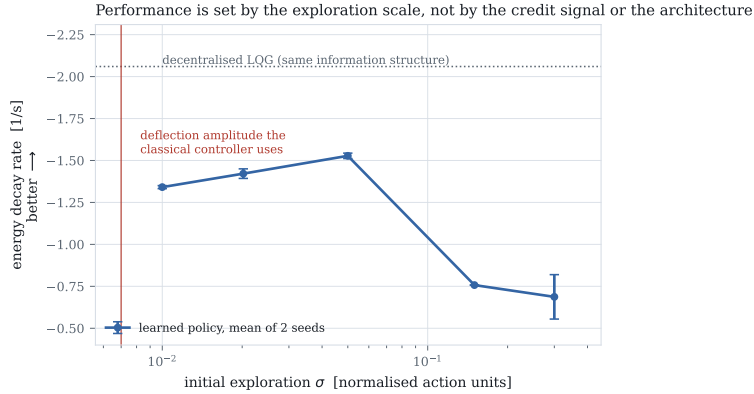


Figure 5: Decay rate against initial exploration standard deviation, all else fixed. The vertical line marks the deflection amplitude the classical controller uses. Two seeds per point.

The policy’s own exploration is then some forty times the amplitude that works, and no credit decomposition can recover a signal buried underneath it.

This is not a tuning remark. Before we found it, every variant plateaued between -0.03 and -0.82 s^{-1} regardless of credit signal, communication, action parameterisation, memory, auxiliary supervision or base controller, and the variant-to-variant differences that motivated several rounds of algorithmic work were of the same order as the seed-to-seed spread — which at $\sigma = 0.30$ was 0.132 and at $\sigma = 0.05$ fell to 0.016. A capability probe rules out the competing explanation: training on the single easiest condition gives $+0.21$ and $+0.27$, worse than training on the full distribution, so the plateau was never a matter of the task being too broad.

The practical recommendation is specific. The useful actuation amplitude is computable before any training, from a classical design or from an estimate of the damping authority required. Scale the exploration to it. On this problem that single number mattered more than every algorithmic choice we made, and it is cheap to check.

11. Discussion

Pulling the pieces together: this study set out to ask whether treating each control surface as a separate learning agent is a workable way to suppress flutter, and the honest answer is conditional. On the credit-assignment question the answer is yes, with numbers behind it. On the question of

whether the resulting policies are ready to fly, the answer is not yet, and the gap to a classical decentralised design is real rather than an artefact of tuning (Section 7). We think both halves of that answer are useful to a reader, which is why neither is relegated to a footnote.

For practitioners. The most actionable finding is the least glamorous one. Before training a policy for a physical control task, compute the actuation amplitude a competent classical controller would use, and check the initial exploration scale against it. On this plant the two were separated by roughly a factor of forty at a conventional PPO setting, and every algorithmic refinement we tried was invisible until that gap was closed. We would expect a version of this check to be worth running on any task where a linear or classical baseline is available even loosely, which covers most physical control problems a control engineering audience would consider.

For the credit-assignment literature. The result in Section 8 is, as far as we can tell, a genuine instance of a difference reward with no estimation step at all, on a task with real physical structure rather than a toy coordination game. That the closed form and the shaping term are individually insufficient but jointly significant is itself informative: it says the myopic, instantaneous part of the credit signal and the long-horizon part are not substitutes for one another on a coupled oscillatory system, which is a more specific claim than “shaping helps.”

On the negative architecture result. We were not expecting the phasor consensus and phase-locked action head to hurt, and reporting that they do is worth more caution than reporting a positive result would need. A plausible interpretation is that a three-surface, six-metre wing with broadcast air data is simply too small a coordination problem for an explicit channel to earn its bandwidth cost in gradient-fitting difficulty, and the eight-surface result — where every condition becomes feasible for the learned policies — is at least consistent with that story, though we did not run it to the statistical standard we hold the main claims to (12). We would read our result as a caution against building communication machinery into a distributed controller by default, on the strength of a physical argument, without first checking whether a plain policy with the same credit signal already does the job.

Why we still find phase coordination the most striking result. A policy trained end to end on a shaped scalar reward, using only its own two accelerometers, converges on the same resonant frequency that a full-state Kalman-filtered

LQG converges on by construction (Section 9). That frequency is not supplied anywhere in the reward, the observation, or the network architecture; it is a property of the coupled structure that the policy has to find. We think this is the sense in which the word “emergent” is earned here, as distinct from being merely asserted.

12. Limitations

- The learned policies do not match the decentralised classical controller on average nominal damping (-1.53 against -2.05 s^{-1}). They are more uniform across the envelope and hold a condition where it diverges; we do not claim they dominate it.
- The robustness result is scale-dependent, as reported in Section 7. We give the trade-off rather than its better half.
- The plant over-estimates control authority by a factor of 1.9 relative to UVLM (Section 4.3). Relative results are unaffected; absolute decay rates would not transfer. We have not retrained on the UVLM plant.
- The credit signal requires the modal control influence $v_r^\top b_k$, free in simulation but requiring a modal model on hardware. Robustness to error in B is untested.
- Proposition 1 concerns the difference reward, not optimality of the resulting policy.
- Ablations use six seeds for the credit comparison and six for the architecture comparison, on a three-surface wing. The eight-surface results are suggestive and were not run to the same standard.
- The exploration finding is established on this plant and this algorithm. We expect it to generalise to control tasks whose useful actuation band is narrow relative to the action range, but have not shown that.

13. Conclusion

Posing flutter suppression as a distributed learning problem exposes a credit structure the physics resolves in closed form: the difference reward is an agent’s own control power, normalised by the instantaneous energy so that it coincides with the objective and cannot be gamed by sustaining the

oscillation. Combining it with log-energy shaping is significantly better than either part alone.

The more transferable lesson is the caution. That result, and a clean negative result on the communication machinery, were both invisible until the exploration scale was matched to the actuation amplitude that damps the wing — a quantity computable in advance from a classical design. On this problem it mattered more than any algorithmic choice we made, and we suspect that is true of active vibration control more broadly.

Code and data availability

The plant, baselines, training code, evaluation harness and the SHARPy cross-validation scripts are available at the project repository.

Appendix A. Testbed parameters

Table A.5 gives the geometric and structural parameters of the wing and Table A.6 the actuator and sensing parameters, common to every controller in this study. All are the values used to generate Table 1.

Table A.5: Wing geometry and structural properties.

Parameter	Value
Semispan	6.096 m
Chord	1.8288 m
Mass per unit length	35.71 kg m ⁻¹
Polar inertia per unit length (about EA)	8.64 kg m
Bending stiffness EI	9.77×10^6 N m ²
Torsional stiffness GJ	0.987×10^6 N m ²
Elastic axis	33 % chord
Centre of gravity	43 % chord
Bending / torsion modes retained	2 / 2
Aerodynamic strips	24
Air density (sea level)	1.225 kg m ⁻³

Appendix B. Training hyperparameters

Table B.7 lists the PPO and network hyperparameters shared across every learned variant in Sections 8 and 10, except where a hyperparameter

Table A.6: Control surface, actuator and sensing parameters (identical for all three surfaces).

Parameter	Value
Spanwise bands (fraction of semispan)	$[0.05, 0.40]$, $[0.40, 0.72]$, $[0.72, 0.98]$
Hinge location	75 % chord
Deflection travel	$\pm 15^\circ$
Rate limit	200° s^{-1}
Actuator time constant	20 ms
Control update rate	200 Hz
Local observation per agent	2 accelerometers, plunge/twist rate, own saturation, air data

is itself the subject of an ablation (the initial exploration standard deviation, and the presence or absence of the architectural mechanisms in Table 4). All runs share parameters across agents, with the agent index supplied as a one-hot input, so the parameter count does not grow with the number of surfaces.

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Table B.7: PPO and network hyperparameters.

Hyperparameter	Value
Parallel environments	128
Rollout length	128 steps
Epochs per update	4
Minibatches per epoch	4
BPTT segment length	32 steps
Learning rate (Adam)	3×10^{-4}
Discount γ	0.995
GAE λ	0.95
PPO clip range	0.2
Value loss coefficient	0.5
Entropy coefficient	2×10^{-3} (0 in the exploration sweep)
Max gradient norm	0.5
Curriculum fraction	0.4 of training
Hidden layer width (trunk)	128
Phasor encoder hidden width	64
Retained modes per phasor	2
Consensus rounds	2
Action-head output-layer gain (init.)	0.01

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